

ZACHARIAS CHEN

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EDUCATION

Duke University

Mechanical Engineering B.S, Computer Science minor | GPA: 3.98/4.00

Durham, NC
8/21 - 5/25

WORK EXPERIENCE

Medtronic

Development Engineer I

Boston, MA
8/25 - Present

- Executed verification activities for software, sustaining capital hardware, and new instrument projects by authoring design verification protocols, analyzing data, evaluating bodies of evidence, and generating verification reports
- Reduced the testing cost for new instrument hand-eye accuracy by at least 10x by decomposing the error into product level requirements using an RSS transfer function, which allows for instrument level verification.
- Downscaled the number of stapling instruments needed for hand-eye accuracy from 21 to 4 through design of statistically rigorous experiments to identify a worst case instrument.
- Developed a bottom-up characterization process for the surgeon console to enable component level verification and mitigate unforeseen design failures.
- Completed green belt training (similar to lean six sigma)

Systems/Development Engineer Intern

6/24 – 8/24

- **Decomposed** Hugo Robotic Assisted Surgery system **requirements** for operator latency, which reduced testing burden by 72-78%, through designing transfer functions and proposing improved testing strategy
- Authored design verification protocol for 11 system-level functionalities through utilization of risk-based design verification and assessment of total body of evidence

Pirouette Medical

Mechanical Design Intern

Portsmouth, NH
5/23 – 8/23

- Increased company completion of epinephrine autoinjector **design verification** by 40% by reviewing requirements, designing experiments, designing and manufacturing fixtures, testing, analyzing data, and making design changes
- Designed test fixtures for CNC milling, injection molding, and SLA printing
- Improved operations by creating assembly fixtures, ordering equipment and software, and improving quality control

Johns Hopkins Applied Physics Lab

Mechanical Engineering Intern

Laurel, MD
8/20 - 5/21

- Researched Capacitive Deionization (CDI) against the current state of art technologies like reverse osmosis, established relationships between electro-chemical properties of materials vs CDI efficiency, ran technoeconomic analysis in Simulink
- Proposed a hybrid desalination pipeline featuring CDI and reverse osmosis to compensate for the shortcomings of both

PUBLICATIONS AND PROJECTS

Brain Tool Laboratory

- **Chen, Z.**, Cahilig, A. C., Dias, S., Kolar, P., Prakash, R., & Codd, P. J. (2025). Where is the Boundary? Multimodal Sensor Fusion Test Bench for Tissue Boundary Delineation. 2025 IEEE 21st International Conference on Body Sensor Networks (BSN), 1–4. <https://doi.org/10.1109/bsn66969.2025.11337667>
- Chen, D., Prakash, R., Wang, V. Y., **Chen, Z.**, Dias, S., Buckland, D. M., Bridgeman, L. J., Oca, S. R., & Mann, B. P. (2026). Design and Evaluation of a Compliant Quasi-Direct Drive End-Effector for Safe Robotic Ultrasound Imaging. IEEE Transactions on Medical Robotics and Bionics, 1–1. <https://doi.org/10.1109/tmr.2026.3699216>
- Prakash, R., Wang, V., **Chen, Z.**, Bridgeman, L., & Codd, P. J. (2025). Portable Dual Sensor Large Area Visualization System for Robotic Laser Surgery. 2025 International Symposium on Medical Robotics (ISMR), 58–65. <https://doi.org/10.1109/ismr67322.2025.11025962>

Duke Aero

Payload Lead (2024-2025)

- '24-25: Led payload development for a 5U CubeSat, overseeing design, manufacturing, integration, and flight operations of guided recovery, gimbal imaging, and computer vision systems
- '23-24: Designed, machined, programmed, and performed ground testing for a 6U CubeSat demonstrating technologies relevant to future planetary colonization and autonomous recovery missions.

Deployable Continuum Robot

- Designed, manufactured, and programmed a Technology Readiness level 3 deployable continuum robot for space inspection applications. Capable of picking up and placing a 250 g object within a 250 mm hemisphere workspace

SKILLS AND INTERESTS

- **Mechanical:** Solidworks, Fusion 360, Onshape, CNC milling, Lathe milling, Wire EDM, FDM and SLA printing, FDM, GDT
- **Programming:** Python, Java, C, C++, MATLAB, SQL, HTML/CSS, MIPs Assembly, PCA, SVM
- **Robotics:** ROS 2, Gazebo, Computer Vision, Forward Kinematics, Sensor Fusion, Path Planning (SLAM, A Star)
- **Verification:** risk-based design verification, design verification, requirements flow down, design of experiments, FEA